

Sales Performance & Profit Analytical Dashboard

Project Objective

- Build an interactive Sales Performance & Profit Analytical Dashboard in Power BI to analyze sales, profit, orders, products, categories, regions, and time-based performance.
- Complete the assignment requirements and add professional features including slicers, drill-down, drill-through, page navigation, reset/bookmark, tooltips, and final validation.

Dataset & Fields

- Main table: SalesData_1000Rows_WithIssues_copy.
- Fields: Order ID, Order Date, Customer Name, Region, Product, Category, Quantity, Unit Price, Sales, Cost, Profit, Profit Cost Difference.
- Supporting tables: Measures and East Region Orders; no relationships were created for these support tables.

Data Cleaning & Loading

- Power Query was used for cleaning and transformation.
- Duplicates were removed. Initial blanks were handled while retaining records where appropriate.
- Order Date was typed as Date and displayed as Long Date. Customer blanks were handled as Unknown. Region blanks were handled as Unknown. Product issues were resolved, including the affected row as Monitor with category Electronics.
- Three Sales nulls were recalculated using $\text{Quantity} \times \text{Unit Price}$.
- One Cost null was completed as 109,859 from $\text{Sales} - \text{Profit}$.
- Two Profit nulls were recalculated as $\text{Sales} - \text{Cost}$.
- Affected cleaned columns replaced the original columns and were renamed back to Sales, Cost, and Profit.
- Final types: Order ID (Whole Number); Order Date (Date); Customer/Region/Product/Category (Text); Quantity/Unit Price/Sales/Cost/Profit/ Profit Cost Difference (Whole Number).

DAX Calculations

- Total Sales = $\text{SUM}(\text{'SalesData_1000Rows_WithIssues_copy'}[\text{Sales}])$
- Total Cost = $\text{SUM}(\text{'SalesData_1000Rows_WithIssues_copy'}[\text{Cost}])$
- Total Profit = $\text{SUM}(\text{'SalesData_1000Rows_WithIssues_copy'}[\text{Profit}])$
- Total Quantity = $\text{SUM}(\text{'SalesData_1000Rows_WithIssues_copy'}[\text{Quantity}])$
- Total Orders = $\text{COUNTROWS}(\text{'SalesData_1000Rows_WithIssues_copy'})$
- Average Order Value = $\text{DIVIDE}([\text{Total Sales}], [\text{Total Orders}], 0)$
- Profit Margin % = $\text{DIVIDE}([\text{Total Profit}], [\text{Total Sales}], 0)$
- Profit Cost Difference calculated column = $\text{Profit} - \text{Cost}$.
- East Region Orders calculated table filters Region = East.
- Required assignment measure Total Profit was created.

Dashboard Structure

- Page 1: Sales Performance Overview

- Page 2: Regional Sales Analysis
- Page 3: Product & Category Analysis
- Page 4: Sales & Profit Trend Analysis
- Page 5: Insights & Recommendations
- Drill-through page: Product Details

Page 1 – Sales Performance Overview

- Title: Sales Performance Overview.
- 5 KPI cards: Total Sales, Total Profit, Profit Margin %, Total Orders, Average Order Value.
- Charts: Sales & Profit by Region (Clustered Column); Order Distribution by Region (Pie – required); Monthly Sales Trend (Line); Sales vs Profit (Line & Clustered Column Combo); Sales by Category (Donut); Regional Sales Performance (Map).
- Map uses Region as Location, Total Sales as Size where available, and Total Profit/Total Orders as tooltips.
- Slicers: Region, Category, Order Date (Between).
- Reset/Clear All button with Reset bookmark is placed on Page 1 only.

Page 2 – Regional Sales Analysis

- Title: Regional Sales Analysis.
- Charts: Profit Margin by Region (Bar); Average Order Value by Region (Column); Category Sales by Region (Stacked Column); Sales Heatmap by Region & Category (Matrix).
- Heatmap: Region rows, Category columns, Total Sales values; Grand Totals off; orange conditional colour scale.
- Slicers: Category and Order Date (Between).

Page 3 – Product & Category Analysis

- Title: Product & Category Analysis.
- Charts: Top 5 Trending Products (required Column); Product Sales Performance (Bar); Product Profit Performance (Bar); Category Profit Analysis (Column); Sales vs Profit by Product (Scatter).
- Top 5 uses Product → Top N → Top 5 → By value Total Orders.
- Scatter: X = Total Sales; Y = Total Profit; Legend = Product; Tooltips = Product, Total Sales, Total Profit; trend line added.
- Drill Down on Category Profit Analysis: Category → Product; tested by selecting Electronics and viewing its products.
- Slicers: Category and Region.

Page 4 – Sales & Profit Trend Analysis

- Title: Sales & Profit Trend Analysis. Performance.
- No KPI cards; page is trend-focused.
- Charts: Profit Trend Over Time (required Line); Monthly Orders Trend (Line); Monthly Profit Margin Trend (Line); Year-wise Sales Performance (Line); Year-wise Profit Performance (Line).
- Order Date Date Hierarchy used where needed.
- Slicers: Region and Order Date (Between).

Page 5 – Insights & Recommendations

- Text-only page with title. No charts and no KPI cards.
- Key Insights: regional sales contribution varies; high-performing products generate stronger order volumes; profitability differs across products/categories based on sales and cost; monthly/yearly trends show changes; order volume identifies stronger demand.
- Recommendations: focus on high-performing products; review pricing/costs for low-profit products; strengthen strategies in lower-performing regions; monitor profit margin; use sales, profit and order trends for decisions.

Slicers & Filtering

- Page 1: Region, Category, Order Date (Between).
- Page 2: Category, Order Date (Between).
- Page 3: Category, Region.
- Page 4: Region, Order Date (Between).
- Page 5: no slicers.

Drill Through

- Separate Product Details page created.
- Drill-through field: Product.
- Back button auto-created and tested.
- Table fields: Product, Category, Region, Total Sales, Total Profit, Total Orders, Total Quantity, Total Cost.
- Right-click Product → Drill through → Product Details was tested successfully.

Page Navigation

- Page Navigator placed on all five main pages.

Reset / Bookmark

- Reset/Clear All button created on Page 1.
- Reset bookmark created for the default filter state.
- Reset behaviour tested: changed filters can be returned to the saved default state.
- Button intentionally kept on Page 1 only.

Tooltips

- Tooltip functionality enabled.
- Category Profit Analysis tooltip includes Total Sales, Total Orders, Total Quantity, and Profit Margin.

Visual Interactions

- Visual interactions were reviewed. Unnecessary cross-filtering was not retained where it did not add value, keeping the dashboard clean and readable.

Final Quality Check

- All five pages reviewed for titles, subtitles, visual alignment, spacing, slicers, navigation, and formatting.
- Required visuals confirmed: regional order distribution Pie, Top 5 Trending Products Column, Profit Trend Over Time Line.

- DAX requirements confirmed: East Region Orders, Profit Cost Difference, Total Profit.
- Slicers, drill-down, drill-through, navigation, reset/bookmark and tooltips confirmed.
- Product Details drill-through and Back navigation tested.

Conclusion

The Sales Performance & Profit Analytical Dashboard combines cleaned sales data, DAX calculations, required assignment visuals, analytical pages, and professional Power BI interactivity into one complete reporting solution suitable for academic submission and portfolio presentation.